

Question 1 Fill in the proof template for **Direct Proofs** below with the appropriate terms.

Claim: $p \rightarrow q$

Proof. Assume $\boxed{p}$

.

.

Thus $\boxed{q}$

Question 2 Choose the book's recommended sequence for the following steps when writing an **if-then** proof.

- Step 1: Unravel the definitions usually found in the assumption and the conclusion of the claim (forward from the beginning of the proof and backward from the end of the proof)
- Step 2: Write the last sentence of the proof (restate the conclusion)
- Step 3: Assume (restate) the hypothesis
- Step 4: Figure out what is missing in linking the two halves of the arguments

Multiple Choice:

- (a) $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$
- (b) $3 \rightarrow 2 \rightarrow 1 \rightarrow 4$ ✓
- (c) $2 \rightarrow 1 \rightarrow 4 \rightarrow 3$
- (d) $3 \rightarrow 2 \rightarrow 4 \rightarrow 1$

Question 3 Which of the following would you need to prove an **if-and-only-if** proof? (Claim: $p \leftrightarrow q$)

Multiple Choice:

- (a) $p \rightarrow q$

- (b) p
- (c) p and q
- (d) $p \rightarrow q$ and $q \rightarrow p$ ✓

(hint: see proof template 2 for if-and-only-if theorem on page 21)

Question 4 In this question, we will walk you through an if-then proof. Fill in the missing steps!

Claim: Suppose a, b, d, x and y are integers. If $d|a$ and $d|b$, then $d|(ax+by)$.

Proof. Assume $d|a$ and $d|b$. [given]

From $d|a$, we can say that $\underbrace{\hspace{2cm}}_{\text{answer1}}$

Similarly, from $d|b$, we can say that $\underbrace{\hspace{2cm}}_{\text{answer2}}$

Then $ax + by = \underbrace{\hspace{2cm}}_{\text{answer3}}$

$= d(mx) + d(ny),$

$= d(mx + ny), \underbrace{\hspace{2cm}}_{\text{answer4}}$

By definition of divides, $\underbrace{\hspace{2cm}}_{\text{answer5}}$

- Answer 1

Multiple Choice:

- (a) $ax = d$
- (b) there exists an integer m , such that $a = dm$ ✓
- (c) $a = dx$

- Answer 2

Multiple Choice:

- (a) there exists an integer m , such that $b = dm$
- (b) there exists an integer n , such that $b = dn$ ✓
- (c) there exists a positive number n , such that $bn = d$
- (d) $b = dy$

- Answer 3

Multiple Choice:

- (a) $(dm)x + (dn)y$ ✓
- (b) $m + n$
- (c) $dx + dy$
- (d) $am + bn$

- Answer 4

Multiple Choice:

- (a) m and n are both positive, $mx + ny$ is also positive
- (b) since m, n, x, y are integers, $mx + ny$ belongs to $\mathbb{Z}$ ✓
- (c) Leave it blank [no step necessary]

- Answer 5

Select All Correct Answers:

- (a) $d|(ax+by)$ ✓
 - (b) $d|(a+b)$
 - (c) $(ax+by)|d$
 - (d) Restate the conclusion part of the if-then claim. ✓
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